

Introduction to Social Data Analytics: Week 9 Part 2

Today's lecture

- ▶ Review last class's application and discuss how to save plots in R
- ▶ Get Out to Vote Experiments
 - ▶ Interpreting regression with binary variables

Back to Florida

- ▶ Let's go back to our butterfly ballot example
- ▶ Additionally, we will discuss how to save plots in R
- ▶ load the data:

```
library(tidyverse)  
florida <- read_csv("florida.csv")
```

Estimate regression predicting Buchanan00 with Perot96

```
lm.florida <- lm(Buchanan00 ~ Perot96, data=florida)
lm.florida

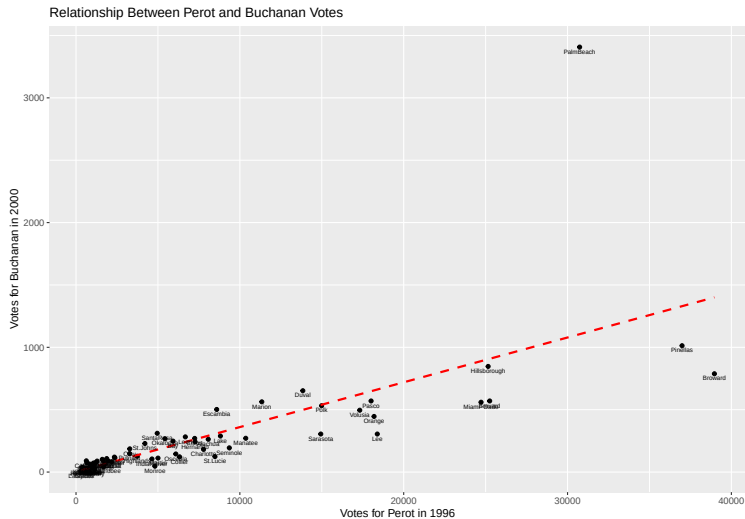
##
## Call:
## lm(formula = Buchanan00 ~ Perot96, data = florida)
##
## Coefficients:
## (Intercept)      Perot96
##      1.34575      0.03592
```

Plot the Regression line

```
# make plot showing relationship between Perot and Buchanan votes  
ggplot(florida, mapping=aes(x=Perot96,y=Buchanan00,label=county)) +  
  geom_point() +  
  geom_smooth(method="lm", se=FALSE, color="red", linetype="dashed") +  
  geom_text(vjust=1.5, size=2) +  
  xlab("Votes for Perot in 1996") +  
  ylab("Votes for Buchanan in 2000") +  
  ggtitle("Relationship Between Perot and Buchanan Votes")
```

Plot the Regression line

```
## `geom_smooth()` using formula = 'y ~ x'
```



How to Save Plots using Code

- ▶ Step 1: tell R what type of file you are creating, and what to name it
- ▶ Step 2: run code to generate the plot
- ▶ Step 3: tell R you are done creating the plot

Step 1: What File Are You Creating

- ▶ In this example, let's create a png file named `florida_scatter.png`. To start, we type

```
png("florida_scatter.png")
```

Step 2: Construct the Plot

```
ggplot(florida, mapping=aes(x=Perot96,y=Buchanan00,label=county)) +  
  geom_point() +  
  geom_smooth(method="lm", se=FALSE, color="red", linetype="dashed") +  
  geom_text(vjust=1.5, size=2) +  
  xlab("Votes for Perot in 1996") +  
  ylab("Votes for Buchanan in 2000") +  
  ggtitle("Relationship Between Perot and Buchanan Votes")
```

Step 3: Close the File

- ▶ The code `dev.off()` tells R that you are done writing to this file
- ▶ So all together, the code to create this file below is:

```
png("florida_scatter.png")

ggplot(florida, mapping=aes(x=Perot96,y=Buchanan00,label=county)) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE, color="red", linetype="dashed") +
  geom_text(vjust=1.5, size=2) +
  xlab("Votes for Perot in 1996") +
  ylab("Votes for Buchanan in 2000") +
  ggtitle("Relationship Between Perot and Buchanan Votes")

dev.off()
```

PDFs instead of PNGs

- ▶ Instead of `png()` you can specify `pdf()` to create a pdf of your plot
 - ▶ Often the image is clearer for pdfs created in this way
- ▶ In our example, all we have to do to create a pdf is change the first line of the code to

Use tidyverse to extract residuals and predicted values

```
florida %>%  
  mutate(predBuchanan00 = lm.florida$fitted.values,  
         residuals=lm.florida$residuals) %>%  
  select(county, predBuchanan00, Buchanan00,  
         residuals) %>%  
  arrange(desc(abs(residuals)))
```

```
## # A tibble: 67 x 4
```

```
##   county      predBuchanan00 Buchanan00 residuals  
##   <chr>          <dbl>          <dbl>    <dbl>  
## 1 PalmBeach      1105.          3407    2302.  
## 2 Broward        1401.           788    -613.  
## 3 Lee            662.           305    -357.  
## 4 Brevard         908.           570    -338.  
## 5 Miami-Dade      889.           560    -329.  
## 6 Pinellas      1330.          1013    -317.  
## 7 Sarasota        538.           305    -233.
```

Is the residual large enough to change the election result?

- ▶ Bush won by 537 votes in Florida
- ▶ Residual is $>2,000$
- ▶ Possible that the butterfly ballot really did decide the election!

Application 2: Primary elections

Puzzle: how can we get people to turn out more in primaries?

- ▶ Primaries really important to selecting candidates
- ▶ California federal general elections 2000-2012: 65.5% turnout among registered voters
- ▶ California federal primary elections 2000-2012: 40.8% turnout among registered voters

Example: Get out the vote experiment

Is lack of mobilization the problem?

- ▶ Hill and Kousser (2016): send mailers to those who vote in general elections in California, but not in primaries encouraging them to vote.

Three randomly assigned arms:

- ▶ Basic info about the election
- ▶ Basic info plus info about the top two primary
- ▶ Basic info plus info on turnout by respondent's party

Election Info Treatment

Dear Registered Voter,

You are currently a registered voter in the State of California. *This letter is to remind you that a Primary Election will be held on Tuesday, June 3, 2014.* Polls will be open from 7 AM to 8 PM on Election Day. Don't forget to vote!

Top-two Treatment

Dear Registered Voter,

You are currently a registered voter in the State of California. *This letter is to remind you that a Primary Election will be held on Tuesday, June 3, 2014.* Polls will be open from 7 AM to 8 PM on Election Day. Don't forget to vote!

All Californians, regardless of party of registration, now have a voice in primary elections. Voters passed Proposition 14, the Top Two Candidates Open Primary Act, in 2010, which changed the primary election process for congressional, statewide, and legislative races. Proposition 14 declared that, "No voter shall be denied the right to vote for the candidate of his or her choice ... based upon his or her disclosure or nondisclosure of party preference." Important changes to be aware of are that:

- All voters can now vote in the primary election.
- All candidates appear on one ballot.
- Candidates are not nominated by the party.
- Any voter can vote for any candidate.
- The two candidates with the most votes move on to the General Election in November.

Partisan Treatment

Dear Registered Voter,

You are currently a registered voter in the State of California. *This letter is to remind you that a Primary Election will be held on Tuesday, June 3, 2014.* Polls will be open from 7 AM to 8 PM on Election Day. Don't forget to vote!

In the last California primary election, held in 2012, only 18.5% of voters like you registered as No Party Preference turned out to vote and make their voices heard. Voting in the primary election determines the candidates for the general election. I hope you will take advantage of this opportunity to make your voice heard.

Scale of Experiment

- ▶ One impressive feature of Hill and Kousser (2016) is the sheer scale of the experiment
 - ▶ In total, the authors sent out about 150,000 mailers
 - ▶ This implies they can detect even small changes in voting behavior
- ▶ While the actual experiment tests between the different treatment arms, we are going to compare individuals who received any type of mailer to those that received no mailer
 - ▶ Voting records are public information, therefore authors had information on whether every individual in California voted or not

Review: Linear Regression

We can write the linear regression model as:

$$Y_i = \beta_0 + \beta_1 \cdot X_i + \epsilon_i$$

- ▶ Y_i : dependent variable
- ▶ X_i : independent variable
- ▶ β_0 : intercept
- ▶ β_1 : slope
- ▶ ϵ_i : error

Review: Binary Variables

Y_i and/or X_i might be binary.

- ▶ X_i might be whether or not a person received a mailer encouraging them to vote.
 - ▶ We will code this as 0 if you did not receive a mailer, 1 if you did receive a mailer
- ▶ Y_i would be whether or not they voted.
 - ▶ We will code this as 0 if you did not vote, 1 if you did vote

Review: Binary Variables

In the case of a binary dependent variable, what would the interpretation of $\hat{\beta}_1$ be?

- ▶ X_i being 1 versus 0 is associated with a $\hat{\beta}_1 \cdot 100$ percentage point change in Y_i
- ▶ For example, receiving a mailer is associated with a $\hat{\beta}_1 \cdot 100$ percentage point change in turning out to vote.

Example

Imagine $\hat{\beta}_0 = 0.2$ and $\hat{\beta}_1 = 0.01$. Then the mailer increases voting by 1 percentage point

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► Predicted Y for **no-mailer**

$$\hat{\beta}_0 + \hat{\beta}_1 \cdot X_i = 0.2 + 0.01 \cdot 0 = 0.2$$

(20 percent vote if **not** sent a mailer)

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- Predicted Y for **no-mailer**

$$\hat{\beta}_0 + \hat{\beta}_1 \cdot X_i = 0.2 + 0.01 \cdot 0 = 0.2$$

(20 percent vote if **not** sent a mailer)

- Predicted Y for **mailer**

$$\hat{\beta}_0 + \hat{\beta}_1 \cdot X_i = 0.2 + 0.01 \cdot 1 = 0.21$$

(21 percent vote if sent a mailer)

Example

Imagine $\hat{\beta}_0 = 0.2$ and $\hat{\beta}_1 = 0.01$. Then the mailer increases voting by 1 percentage point

- Predicted Y for **no-mailer**

$$\hat{\beta}_0 + \hat{\beta}_1 \cdot X_i = 0.2 + 0.01 \cdot 0 = 0.2$$

(20 percent vote if **not** sent a mailer)

- Predicted Y for **mailer**

$$\hat{\beta}_0 + \hat{\beta}_1 \cdot X_i = 0.2 + 0.01 \cdot 1 = 0.21$$

(21 percent vote if sent a mailer)

Difference in voting rates is 1 percentage point or (i.e. $\hat{\beta}_1 \cdot 100$)

Load the data and explore experimental conditions

```
library(tidyverse)
turnout <- read_csv("turnout.csv")
```

► What values does the variable `treatment.assign` take:

```
table(turnout$treatment.assign)
```

```
##
##      Control Election info      Partisan Top-two info
##      3722672          29885          59857          59854
```

Estimating the impact of mailer

- ▶ Our dependent variable will be `yvar`
 - ▶ This variable is equal to 1 if the individual voted and zero otherwise
- ▶ Our independent variable will be `mailer`
 - ▶ This is equal to 1 if the individual is in any of the following treatments: Election Info, Partisan, or Top-two info

Estimate effect of receiving any mailer

```
#Run regression
```

```
lm.turnout <- lm(yvar ~ mailer, data=turnout)
```

```
lm.turnout
```

```
##
```

```
## Call:
```

```
## lm(formula = yvar ~ mailer, data = turnout)
```

```
##
```

```
## Coefficients:
```

```
## (Intercept)      mailer
```

```
##      0.093125      0.004899
```

Interpretation

- ▶ Receiving a mailer increased turnout by 0.49 percentage points.
- ▶ Without a mailer, 9.3% of those in the sample turned out to vote in the primary.
- ▶ With a mailer, 9.8% of the sample turned out to vote.

Heterogeneity

- ▶ Let's see if the treatment effect varies by the party of the individual. In other words are Democrats or Republicans more likely to be influenced to vote by this mailer?
- ▶ To do so, let's estimate the same regression, but for Democrats and Republicans separately
- ▶ As an initial step, let's construct dataframes restricted to Republicans, and then another restricted to Democrats

```
df_rep <- turnout %>% filter(Party=="REP")  
df_dem <- turnout %>% filter(Party=="DEM")
```

Estimate Regression separately for each group

```
lm.rep <- lm(yvar ~ mailer, data = df_rep)
lm.dem <- lm(yvar ~ mailer, data = df_dem)
```

► Results for Republicans

```
lm.rep
```

```
##
## Call:
## lm(formula = yvar ~ mailer, data = df_rep)
##
## Coefficients:
## (Intercept)      mailer
##    0.106132    0.006654
```

Results for Democrats

```
lm.dem
```

```
##  
## Call:  
## lm(formula = yvar ~ mailer, data = df_dem)  
##  
## Coefficients:  
## (Intercept)      mailer  
##    0.090181    0.003961
```

- Overall, the treatment effect is slightly higher for Republicans (0.007) relative to Democrats (0.004)